Software Requirements Specification
SweetLedger

SWEETLEDGER

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a clear and detailed description of the Sweetledger system for all project stakeholders. This includes the system users, the client (Dr. Ermias Mamo), and the development team. The SRS serves as a reference point for current and future clarification regarding the system's functionality, design, and scope.

This document will be maintained and updated throughout the project lifecycle and reviewed during each of the five sprint presentations conducted over the course of the semester. The requirements outlined here have been approved and must not be altered without formal authorization to ensure the successful and consistent completion of the system.

1.2 Scope

Sweetledger is a web-based accounting ledger system designed to simulate a professional-grade accounting platform suitable for company use. The project is divided into five sprints, with each sprint building upon the previous one to incrementally develop a complete and functional system.

The client, Dr. Ermias Mamo, will evaluate each sprint deliverable to ensure compliance with the project's requirements and objectives. Sweetledger will utilize Cloudflare as the hosting platform and MongoDB as the database. Cloudflare Pages is used for the frontend, whilst Render is used for the backend database. The primary implementation languages include JavaScript, CSS, and HTML.

1.3 Definitions, Abbreviations, and Acronyms

Acronym	Meaning
UI	User Interface
COA	Chart of Accounts
SRS	Software Requirements Specification
DB	Database
PR	Post Reference

1.4 Reference Material

"Getting Started." *React.dev*, 17 Jul. 2019, react.dev/learn/getting-started.

MongoDB. "Getting Started." *MongoDB Documentation*, MongoDB, 1 Oct. 2025, www.mongodb.com/docs/get-started/. Accessed 1 Oct. 2025.

Institute of Electrical and Electronics Engineers. *IEEE Recommended Practice for Software Requirements Specifications.* IEEE Std 830-1998, IEEE, 1998.

2. Overall Description

This section provides an overview of the rationale and structure behind the system requirements for Sweetledger. The project is divided into the following sprints:

- **Sprint 1:** User Interface and Login Module
    - Focuses on the design and deployment of Sweetledger's user interface, including authentication features such as login and account creation.

- **Sprint 2:** Chart of Accounts Module
  - Defines user roles and access levels, and manages the setup and display of the chart of accounts.
- **Sprint 3:** Journalizing and Ledger Module
  - Handles the creation, storage, and viewing of journal entries and general ledger accounts.
- **Sprint 4:** Adjusting Entries and Financial Reports Module
  - Implements adjusting entries functionality and enables the generation of financial reports, such as the Trial Balance, Income Statement, and Balance Sheet.
- **Sprint 5:** Financial Reports & Dashboard
  - Integrates all system components into a unified dashboard that displays key financial ratios, performance indicators, and system notifications. This module serves as the landing page for all user roles, offering visual insights into the organization's financial health and overall system activity.

3. Performance Requirements and Reliability

Sweetledger is designed to be both efficient and reliable. The system's performance will benefit from the technologies chosen—Cloudflare and MongoDB—which ensure fast response times, scalability, and stability.

Although Cloudflare may experience brief initialization delays after extended downtime, it provides highly reliable uptime once active. MongoDB's integration with Amazon Web Services supports strong data consistency and reduces potential system failures through streamlined architecture.

A mobile version of Sweetledger will also be developed during the final stages of deployment to enhance accessibility. By leveraging these technologies and applying sound software engineering practices, Sweetledger aims to deliver a smooth, responsive, and dependable user experience across all modules.

4. System Features

4.1 User Interface Module

4.1.1 Description

This high-priority feature provides the authentication and user management interface for Sweetledger. It allows administrators, managers, and regular users (accountants) to log in, manage accounts securely, and access system functionalities based on assigned roles. The module enforces strong password policies, user activation control, and system-wide consistency in layout and accessibility.

4.1.2 Stimulus/Response Sequences

- **User Action:** Navigates to the login page.

  System Response: Displays login form containing username and password fields, submit button, “Forgot Password,” “Create New User,” and the system logo.
- **User Action:** Enters credentials and clicks “Submit.”

  System Response: Authenticates credentials, displays the user dashboard with the username and profile picture on the top-right corner.
- **User Action:** Fails login three consecutive times.

  System Response: Temporarily suspends account and displays suspension notice.
- **User Action:** Clicks “Forgot Password.”

  System Response: Prompts user to enter email and user ID, asks security questions, and allows password reset if verified.
- **User Action:** Clicks “Create New User.”

  System Response: Displays a registration form for first-time access (first name, last name, address, DOB). Sends requests to the administrator for approval.
- **Administrator Action:** Approves or rejects account request via email notification.

  System Response: Sends appropriate email notification to requester with login instructions or rejection message.

4.1.3 Functional Requirements

- **ACM-01:** The system shall support three roles: Administrator, Manager, and Regular User (Accountant).
- **ACM-02:** The system shall authenticate users and redirect them to dashboards based on their role.
- **UI-01:** The login page shall contain:
    - Username textbox
    - Password textbox (encrypt input)
    - Submit button
    - Forgot Password button
    - Create New User button
    - Company logo (visible on all pages)
- **ACM-03:** The system shall allow first-time users to submit registration requests, capturing: First Name, Last Name, Address, Date of Birth
- **ACM-04:** Upon submission, an email shall be sent to the administrator for approval. If approved, the system shall send the user a login link via email.
- **SEC-06:** The system shall require users to verify their identity using email, user ID, and security questions before resetting a password.
- **UI-02:** Upon login, the username and profile picture shall appear on the top-right corner of all pages, and the company logo shall remain visible throughout the interface.
- **ADM-01:** Administrators shall have access to reports of all system users, including account status, activation, and expiration details.

- **ADM-02:** Administrators shall be able to suspend users for specific date ranges (e.g., leave of absence).
- **ADM-03:** Administrators shall be able to send system emails directly to any user.
- **ACM-05:** Usernames shall follow the pattern: <FirstInitial><FullLastName><MMYY>
- **DBM-01:** All login attempts, password histories, and user activities shall be stored securely in database tables.

4.2 Chart of Accounts Module

4.2.1 Description

This high-priority feature allows administrators to create and manage the Chart of Accounts, including detailed account information and balances. The module supports structured categorization, validation of numeric account codes, prevention of duplicates, and real-time event logging. Managers and accountants can view accounts but cannot modify or deactivate them.

4.2.2 Stimulus/Response Sequences

- **Administrator Action:** Opens "Chart of Accounts" page.
  System Response: Displays all existing accounts with options to Add, Edit, View, or Deactivate.

- **Administrator Action:** Adds a new account.

  System Response: Prompts for account details (name, number, description, category, etc.), validates uniqueness, and saves data.
- **Administrator Action:** Attempts to add a duplicate account number or name.

  System Response: Displays an error message and prevents the save operation.
- **Administrator Action:** Views an account record.

  System Response: Displays full account details, including balance, type, and creation date.
- **Administrator Action:** Deactivates an account with zero balance.

  System Response: Marks the account as inactive and logs the change.
- **Administrator Action:** Attempts to deactivate an account with a non-zero balance.

  System Response: Displays an error message preventing the action.
- **Manager/Accountant Action:** Views the Chart of Accounts or searches for an account by number or name.

  System Response: Displays filtered results without edit permissions.
- **User Action:** Administrator adds, edits, views, or deactivates an account.

  System Response: Updates the chart of accounts and logs the action with before/after images (if applicable), user ID, date, and time.

4.2.3 Functional Requirements

- **COA-1:** The system shall allow administrators to Add, View, Edit, and Deactivate accounts.
- **COA-02:** Each account shall include:
  - Account Name
  - Account Number (numeric only, no decimals or letters)
  - Description
  - Normal Side (Debit/Credit)
  - Category (Asset, Liability, Equity, Revenue, Expense)
  - Subcategory (e.g., Current Assets)
  - Initial Balance
  - Debit and Credit fields
  - Calculated Balance
  - Date/Time Added
  - User ID of the creator
  - Order (e.g., 01 for Cash)
  - Statement Type (IS, BS, RE)
  - Comment
- **COA-03:** The system shall not allow duplicate account names or numbers. Account numbers shall not allow decimal spaces or alphanumeric values.
- **COA-04:** All monetary values shall display with two decimal places and commas when appropriate.
- **COA-05:** Accounts with balances greater than zero cannot be deactivated.

- **COA-06:** Adim should be able to view individual account details and generate full Chart of Accounts reports
- **COA-07:** The system shall provide search and filter capabilities by account name, number, category, subcategory, and amount.
- **UI-03:**
  - The logged-in username shall appear in the top-left corner.
  - The company logo shall appear on all pages.
  - The interface shall maintain consistent color and layout.
  - Buttons shall have tooltips describing their function.
  - A “Help” button shall be present on each page.
- **COA-08:** Clicking an account shall navigate to that account’s detailed ledger.
- **EVL-01:** For every record added, modified, or deactivated:
  - The system shall generate an event log showing before and after images.
  - Logs shall include User ID, Date, Time, and unique Event ID.
  - View event logs with before/after images, user ID, and timestamp.
- **DBM-02**: Event Logs for each record shall be stored in the database
- **COA-09:** Managers and accountants shall be able to view all accounts, but cannot add, edit, or deactivate them.
- **UI-04:** A pop-up calendar shall display at the top-left corner of the Chart of Accounts page. Navigation buttons for other modules (e.g., Journal Entries, Reports) shall appear at the top of each page.

4.3 Journalizing & Ledger Module

4.3.1 Description

This feature allows administrators, managers, and accountants to manage accounts, create and approve journal entries, and maintain accurate ledgers. The module ensures proper validation of transactions, role-based access, audit trails through event logs, and communication capabilities. It supports source document attachment, error handling, and full traceability from journal entry to ledger posting.

4.3.2 Stimulus/Response Sequences

- **User Action**: Administrator clicks to send an email to a manager or accountant

  System Response: System opens the email interface from the chart of accounts page and sends a message.

- **User Action**: Manager creates a journal entry.

  System Response: System validates accounts, debits, credits, and allows submission.

- **User Action**: Manager approves a journal entry.

  System Response: Journal entry status changes to “Approved,” and posting reflects in the ledger.

- **User Action**: User clicks Post Reference in the ledger.

  System Response: System navigates to the originating journal entry.

- **User Action**: Accountant clicks the account name in the chart of accounts.

  System Response: System navigates to the ledger page for that account.

4.3.3 Functional Requirements

- **COA-10:** Users can send emails to manager or accountant users from the chart of accounts page.
- **JOR-01:** Managers & accountants can create new journal entries only using accounts in the chart of accounts.
- **JOR-02**: Debits must come before credits. Multiple debits and credits per entry are allowed.
- **JOR-03**: Entries can be canceled or reset before submission. Once submitted, entries cannot be deleted.
- **JOR-04**: Each journal entry must have at least one debit and one credit.
- **JOR-05:** For entry to be valid, debits must equal credits.
- **JOR-06**: Source documents of type PDF, Word, Excel, CSV, JPG, and PNG can be attached to new journal entries.
- **JOR-07:** The Manager can approve or reject entries. Rejection requires a reason in a comment field.
  - **JOR-08**: Once entry is approved, it must be reflected in the ledger automatically.

- **JOR-09**: Must be able to filter entries based on status (pending, approved, rejected) by date
- **JOR-10**: Must be able to view entries created by other managers or accountants
- **LED-01:** Clicking an account number in a journal entry navigates to its ledger.
- **LED-02**: Ledger entries display: date, description, debit, credit, and balance.
- **LED-03**: Each ledger entry includes a clickable Post Reference to the originating journal entry.
- **LED-04**: Filtering and searching by date, date range, account name, or amount is required.
- **LED-05**: Ledger balances must be accurate after each posting.
- **ERR-01:** Display error messages in red.
- **JOR-11**: Errors such as unequal debit and credits prevent the submission of invalid entries.
- **ERR-02** Log all errors in a database table.
    - Examples: unbalanced debits/credits, missing source documents, invalid accounts.
- **EVL-01:** All roles can view event logs for accounts. Logs show before and after images, user ID, date, and time of change.
- **JOR-12:** The system notifies managers when journal entries are submitted for approval.

4.4 Financial Reports Module

4.4.1 Description

This feature supports the generation of financial statements and ensures traceability between journals, ledgers, and financial reports.

4.4.2 Stimulus/Response Sequences

- **User Action:** User generates a financial report (Trial Balance, Income Statement, Balance Sheet, Retained Earnings Statement).
  System Response: System produces the report for the specified date or range, with options to view, save, print, or email.

4.4.3 Functional Requirements

- **REP-01:** Generate, view, save, email, or print Trial Balance, Income Statement, Balance Sheet, and Retained Earnings Statement for a specific date or range.

4.5 Financial Ratios & Dashboard Module

4.5.1 Description

This module provides users with a dynamic dashboard displaying key financial ratios and summary insights upon login. It consolidates accounting data from journals and ledgers into visual indicators for quick assessment of the organization's financial health. The dashboard serves as the system's landing page, ensuring each user role—Administrator, Manager, and Accountant—can access relevant information, notifications, and navigation options. The interface maintains consistency with the overall application design and supports color-coded ratio evaluation for intuitive understanding.

4.5.2 Stimulus/Response Sequences

- **User Action:** The user logs into the system and accesses the landing page (dashboard).

  System Response: The landing page loads the appropriate navigation buttons based on user role (Administrator, Manager, Accountant) and shows validated financial data and calculates ratios.
- **User Action**: The user clicks View System Notifications.

  System Response: System messages or alerts (e.g., pending journal approvals) are displayed in the notification area.

- **User Action**: The user views financial ratios and related performance indicators.

  System Response: The dashboard displays ratios with color-coded performance indicators: Green, Yellow, Red

4.5.3 Functional Requirements

- **FIN-01**: All users shall be able to view key financial ratios on the landing page once successfully logged into the system

- **FIN-02**: The dashboard shall display each financial ratio using a three-color scheme to indicate performance:
    - **Green** – Good
    - **Yellow** – Borderline or warning range
    - **Red** – Needs attention or further review
        - Reference ranges for each ratio shall be based on standard financial benchmarks.
- **DSH-01**: The landing page shall serve as the user’s primary dashboard after login.
- **DSH-02**: Menu buttons and dashboard components visible on the landing page shall correspond to each user’s role permissions.
- **DSH-03**: A section of the landing page shall display important system messages, such as:
    - Journal entries pending approval
    - System alerts or announcements
    - Upcoming deadlines or account status notifications
- **DSH-04**: The landing page’s layout, color scheme, and design elements shall remain visually consistent with all other pages in the application.

5. Non-Functional Requirements

5.1. Security Requirements

5.1.1. **SEC-01:** All users shall be required to log in using secure credentials that are encrypted and stored in the database

5.1.2. **SEC-02:** Passwords must meet complexity rules (minimum 8 characters, include letters, numbers, and special characters, and start with a letter).

5.1.3. **SEC-03:** Only authorized users with appropriate roles (Administrator, Manager, Accountant) shall be able to access restricted modules such as User Management and Chart of Accounts modification.

5.1.4. **SEC-04:** The system shall automatically suspend a user account after three consecutive failed login attempts.

5.1.5. **SEC-05:** The system shall notify users by email three days before password expiration and prevent the reuse of previously used passwords

5.2. Usability Requirements

5.2.1. **USE-01:** The system shall display confirmation or error messages within two seconds of any major user-triggered action, such as login, user creation, or account modification.

5.2.2. **USE-02:** All user interface elements shall be clearly labeled, with tooltips explaining button functionality and a consistent color scheme for all modules.

5.2.3. **USE-03:** A built-in “Help” button shall be accessible from every page, providing context-based help topics for users.

5.2.4. **USE-04:** The system shall allow users (Administrator, Manager, Accountant) to log in and access their dashboards with no more than three clicks from the homepage.

5.2.5. **USE-05**: Users must easily navigate from the chart of accounts → ledger → journal entry (clickable links).

5.3. Compatibility Requirements

5.3.1. **COMP-01:** The application shall support standard accounting data exchange formats (e.g., CSV) for import and export functionality.

5.3.2. **COMP-02:** The application shall support mobile device access.

5.3.3. **COMP-03:** The application shall support Mac, Windows, and Linux environments.